15 **C** As temperature is constant, the internal energy of the gas is unchanged. Hence A is wrong.

Work is done by the gas during expansion, hence B is wrong.

C is correct as $\Delta U = Q + W \Rightarrow Q = -W$ when $\Delta U = 0$

D is wrong as temperature is unchanged thus root-mean-square speed is unchanged.

16 C Positions of the KE = 0 (*at t = 0, T/2 and T*) corresponds to x = amplitude